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ENGINEERING



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Smart Automatic LED Lighting System And IR Sensor using Python

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introduction

- **Smart lighting systems** are designed to automate the control of lights, reducing the need for manual switching and improving user convenience.
- **Energy conservation** has become an important requirement in homes, offices, classrooms, and public places to reduce electricity consumption and operating costs.
- This project uses an **Infrared (IR) sensor** to detect the presence or movement of a person within a specific area.
- the **LED lights automatically turn ON**, and when no movement is detected, the lights turn OFF after a certain period.



Objectives

❑ Project Goal

To develop a smart automatic LED lighting system that detects human presence using an IR sensor and controls lighting automatically to save energy.

❑ Technology Used

Python, IR Sensor, LED Module, Arduino/Raspberry Pi, and Serial Communication for real-time detection and lighting control.

❑ Expected Impact

The system reduces electricity wastage, improves energy efficiency, lowers power costs, and provides a convenient automated lighting solution.

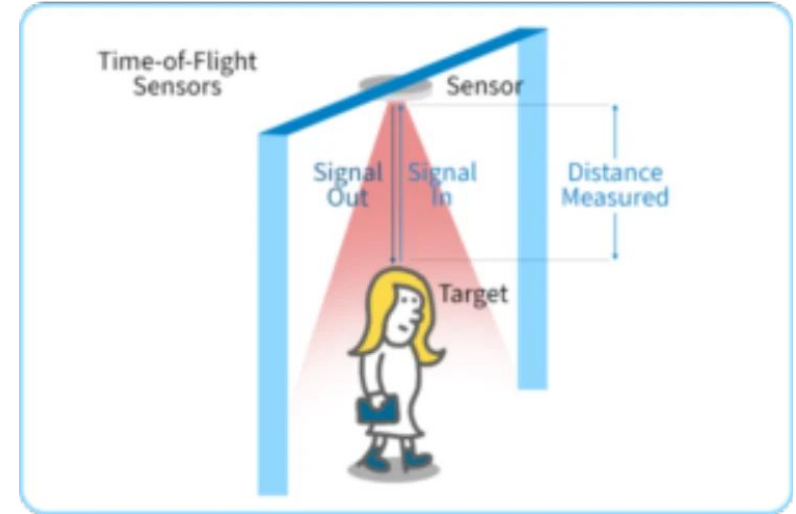


Literature Survey / Existing System

Author & Year	Existing System	Limitations
A. Kumar & P. Sharma (2024)	Manual Lighting Control System	requires manual switching of lights. Users may forget to turn OFF lights, causing energy wastage.
A. Kumar & P. Sharma (2024)	Timer-Based Automatic Lighting System	Uses fixed timers and cannot detect actual human presence. Lights may remain ON unnecessarily.
M. Singh & K. Rao (2025)	Mobile App Controlled Smart Lighting	Allows remote control through mobile apps but depends on user interaction and internet connectivity

Problem Statement

- **Context:** Lighting systems are commonly used in homes, classrooms, and offices for illumination. Most traditional lighting systems require manual operation.
- **Problem:** Lights are often left ON when no one is present, causing unnecessary power consumption and higher electricity bills.
- **Why It Matters:** An automatic LED lighting system using an IR sensor helps save energy, reduce electricity costs, and improve convenience by controlling lights automatically.



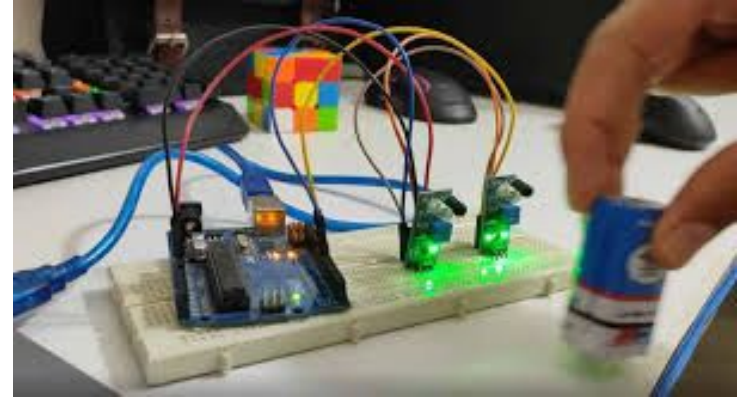
Module-1: person counting and monitoring module

- ❑ Counts the number of people entering and leaving the area.
- ❑ Maintains a real-time occupancy count.
- ❑ Receives detection signals from IR sensors.
- ❑ Updates the person count automatically.
- ❑ Displays the current number of occupants.
- ❑ Helps monitor room usage efficiently.
- ❑ Supports energy-saving by enabling occupancy-based lighting control

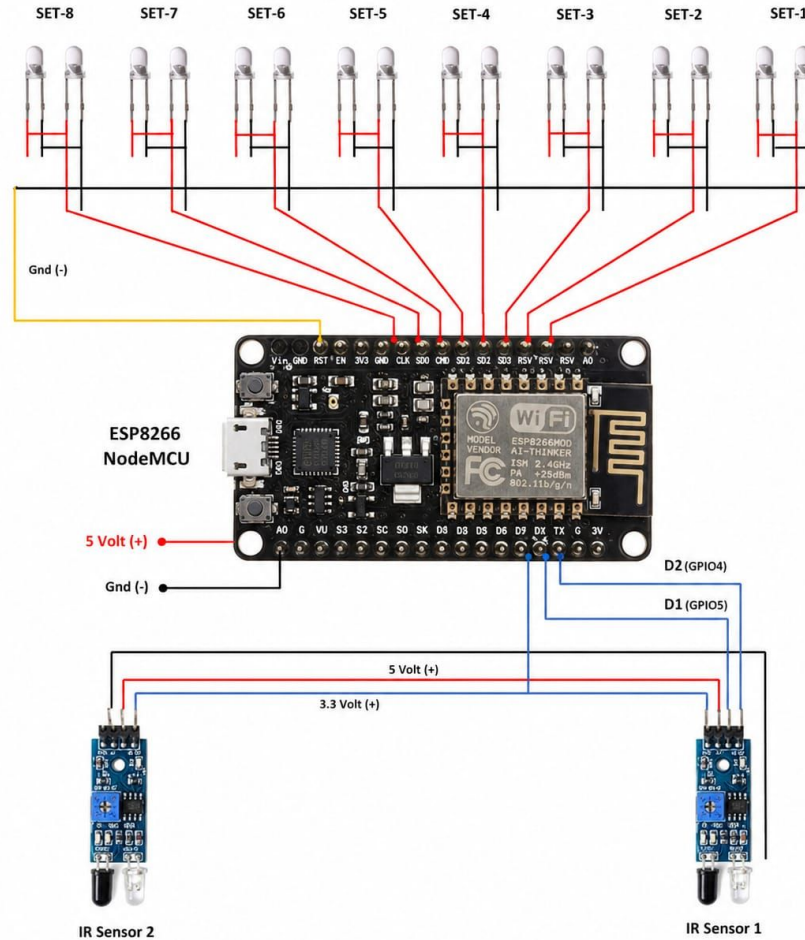


Module-2:IR sensor-based person detection

- ❑ Uses IR sensors to detect human presence and movement.
- ❑ Continuously monitors the entry and exit points.
- ❑ Detects interruptions in the infrared beam.
- ❑ Sends detection signals to the Python program.
- ❑ Provides fast and accurate person detection. Operates automatically without manual intervention.
- ❑ Forms the core sensing unit of the smart lighting system.



Staircase Light Circuit Diagram



Tools and Technologies

- **Front-end**

A user-friendly website interface developed using **HTML, CSS, and JavaScript** to display sensor status, LED status, and system monitoring information.

- **Back-end**

Python (Flask) is used to process sensor inputs, control LED operations, and manage communication between hardware and the website.

- **Hardware Components**

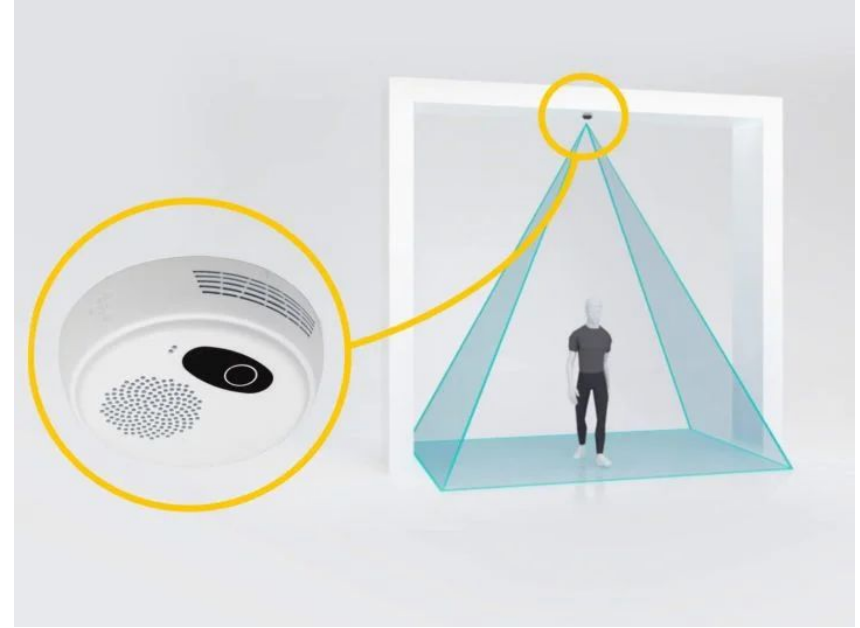
IR Sensor, LED Module, Arduino Uno / Raspberry Pi, Breadboard and Jumper Wires

- **Tech Stack**

Python, Flask, HTML, CSS, JavaScript, IR Sensor Technology, and Arduino/Raspberry Pi for real-time automatic lighting control and monitoring.

Conclusion

- Developed a smart automatic LED lighting system using an IR sensor and Python.
- The system detects human presence and controls LED lights automatically.
- It reduces unnecessary power consumption and saves electricity.
- The project provides a simple, low-cost, and efficient lighting solution.
- Real-time operation improves user convenience and energy management.
- The system can be further enhanced with IoT and mobile monitoring features.



References

- A. Rajasekar and S. Vijayalakshmi, *"Design and Implementation of Automatic Lighting Control System Using IR Sensors,"* International Journal of Engineering Research and Technology (IJERT), vol. 8, no. 6, pp. 245–250, 2019.
- B. M. Patel and R. Shah, *"Energy Efficient Smart Lighting System Using Sensors and Microcontrollers,"* International Conference on Smart Technologies and Systems for Next Generation Computing, pp. 112–117, 2020.
- C. Python Software Foundation, *"Python Documentation,"* Available: <https://www.python.org/doc/>
- D. Arduino Documentation Team, *"Arduino Uno Rev3 Technical Specifications,"* Available: <https://www.arduino.cc/>
- E. E. Gomez and J. Martin, *"IoT-Based Smart Home Lighting Automation Using Sensors,"* International Journal of Smart Home Applications, vol. 12, no. 3, pp. 45–52, 2021. Flask Development Team,
- F. *"Flask Web Framework Documentation,"* Available: <https://flask.palletsprojects.com/>



thank you